

LA HW

Included exercises: 2, 4-5, 8, 37-38, 45, 47, 53

Exercises 1-8

In Exercises 1–8, determine whether the given matrix is orthogonal.

$$A = \begin{bmatrix} 1 & 5 & 2 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Original Exercise 2

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Original Exercise 4

$$I_5$$

Original Exercise 5

$$\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$$

Original Exercise 8

$$\left| \begin{array}{ccc} \frac{2}{3} & \frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} \\ \frac{2}{3} & -\frac{\sqrt{2}}{2} & \frac{\sqrt{2}}{6} \end{array} \right|$$

Original Exercise 37

Find an orthogonal operator T on $\mathcal{R}^3$ such that

$$T \left(\frac{1}{7} \begin{bmatrix} 3 \\ -2 \end{bmatrix} \right) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Original Exercise 38

$$\begin{bmatrix} 1 \\ 6 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Find an orthogonal operator T on $\mathcal{R}^3$ such that $T(\mathbf{v}) = \mathbf{w}$, where

$$\mathbf{v} = \frac{1}{\sqrt{10}} \begin{bmatrix} 3 \\ 1 \\ 0 \end{bmatrix} \quad \text{and} \quad \mathbf{w} = \frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ -2 \\ 1 \end{bmatrix}.$$

Original Exercise 45

for W .

Let $\{\mathbf{v}, \mathbf{w}\}$ be an orthonormal basis for $\mathcal{R}^2$, and let $T: \mathcal{R}^2 \rightarrow \mathcal{R}^2$ be the function defined by

$$T(\mathbf{u}) = (\mathbf{u} \cdot \mathbf{v} \cos \theta + \mathbf{u} \cdot \mathbf{w} \sin \theta) \mathbf{v} \\ + (-\mathbf{u} \cdot \mathbf{v} \sin \theta + \mathbf{u} \cdot \mathbf{w} \cos \theta) \mathbf{w}.$$

Original Exercise 47

the dot product of the i th and j th rows of Q .

Use Theorem 6.10 to prove that if T and U are orthogonal operators on $\mathcal{R}^n$, then both TU and T^{-1} are orthogonal

Original Exercise 53

$\theta/2$, then TU is the reflection about $\mathcal{L}'$.

Let W be a 1-dimensional subspace of $\mathcal{R}^2$. Regard W as a line containing the origin. Let P_W be the orthogonal projection matrix on W , and let $Q_W = 2P_W - I_2$. Prove the following results:

- (a) $Q_W^T = Q_W$.
- (b) $Q_W^2 = I_2$.
- (c) Q_W is an orthogonal matrix.